Mixed-Unit Refresher

Name: _____ Date: _____

5 questions | 100 points | About 40 minutes | Four-function calculator

Show your work and explain your reasoning. Give exact answers.

1. **20 points**

Let R be the function below.

$$R(x) = \frac{4x^2 + 6x + 2}{6x^2 + 15x + 6}$$

(a) Simplify and state the original domain in interval notation.

Work

(b) Find all intercepts, any holes (coordinates), and all vertical and horizontal asymptotes. Show your calculations.

Work

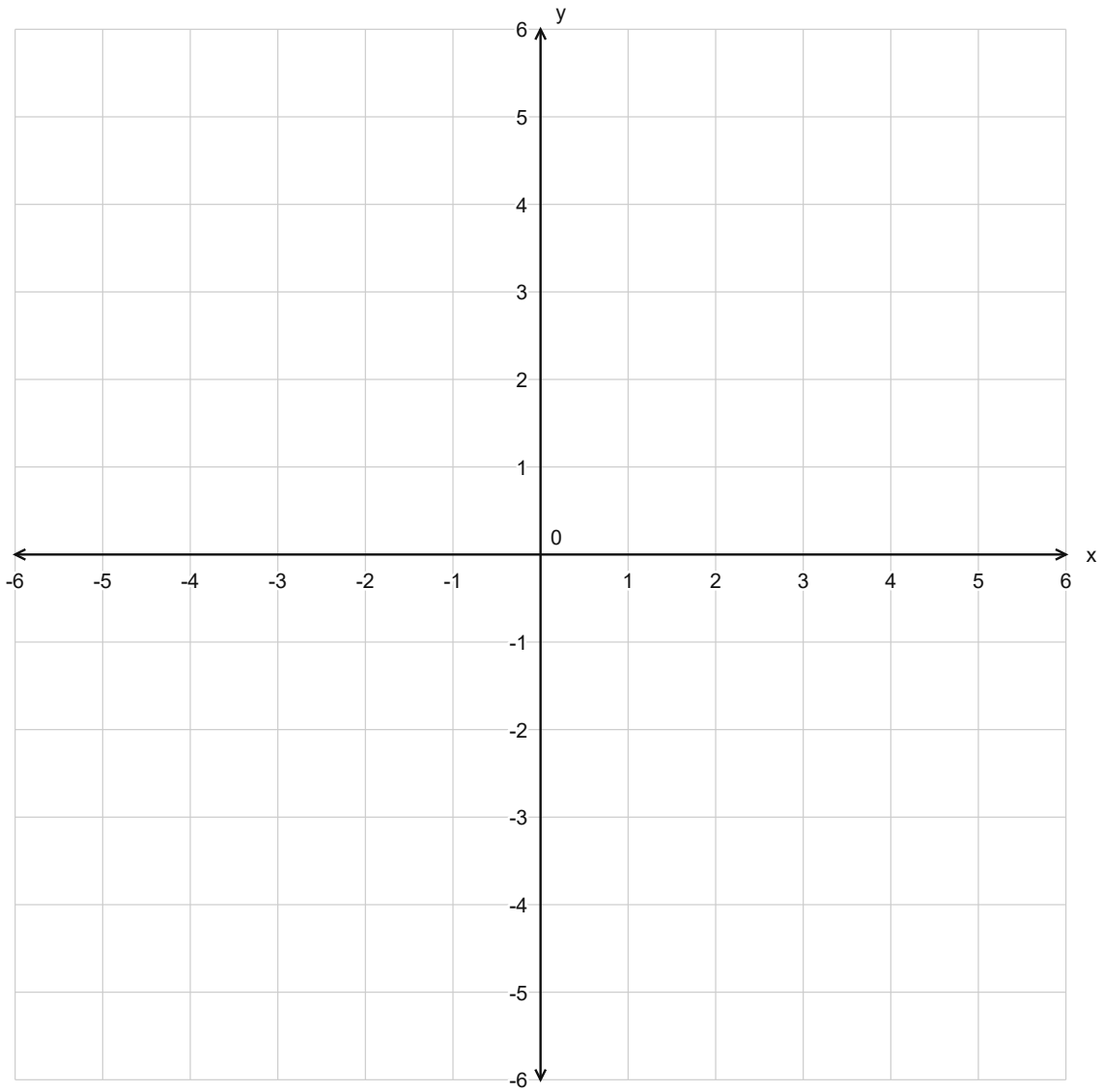
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1. (continued)

(c) Sketch R on the grid. Label the features you found in part (b).

Each square represents 1 unit on both axes.



Explain what happens as x approaches the vertical asymptote from each side.

Answer

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2. **20 points**

Let S be the function below.

$$S(x) = \frac{6x^3 - 7x^2 - 2x + 2}{3x^2 + x - 2}$$

(a) Find the slant asymptote using polynomial long division. Show all work.

Work

(b) Describe the far-left and far-right end behavior. Explain algebraically why your line describes that behavior.

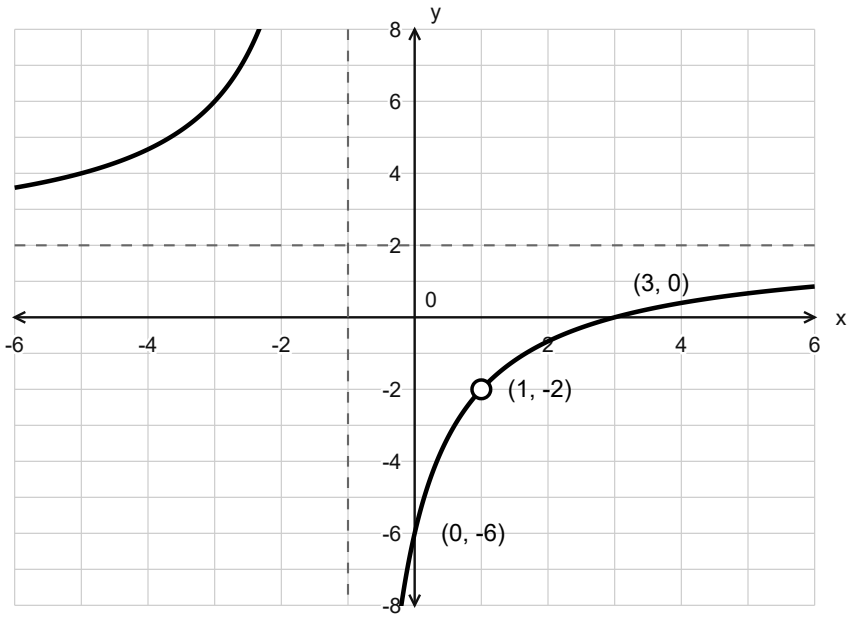
Answer

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3. **20 points**

The graph continues beyond the window. Dashed lines are asymptotes.
The open circle is its only missing point away from the vertical asymptote.



(a) State the domain in interval notation.

(b) Write one possible rational function that matches the graph, including its hole. Explain how it matches the intercepts, asymptotes, and hole.

Work

(c) Describe the far-left and far-right end behavior.

Answer

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4. **20 points**

A printer takes a fixed setup time plus a constant time per page. $T(p)$ is the total job time in seconds for p pages. Selected values are shown.

p **(pages)**	2	4	6	8	10	12
$T(p)$ **(sec)**	8	13	18	23	28	33

(a) Find $T(8)$ and $T^{-1}(8)$. Explain what each means, including units.

Answer

(b) Write formulas for T and T^{-1}. Show your algebra.

Work

(c) Use your inverse to determine the number of pages in a 53-second job.

Work

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5. **20 points**

Let U be the function below.

$$U(x) = 2 + \sqrt{5x + 10}$$

(a) Find U^{-1}. State the domains of U and U^{-1}. Show your algebra.

Work

(b) What should each expression below equal if the functions are inverses? Write your answers first. Then verify each with algebra and state the allowed x-values.

$U(U^{-1}(x))$ **=** _____

$U^{-1}(U(x))$ **=** _____